Final Design Report Template

Maximum Length: 10 pages (double column, figures included, references excluded)
Font: Times New Roman
Columns: 2-column IEEE style
Spacing: Single
Figures/Tables: Label and captioned
File Name: TeamName_FinalReport.pdf

TITLE (Centered, Bold, 16pt)

Subtitle if needed

Team Name
Member Names (IDs)
Department/Faculty
University of the West Indies

Abstract (150–200 words)

Summarize:

- Problem
- Proposed robotic solution
- Key design features
- Main results/benefits
- Sustainability impact

Write one concise paragraph only.

Introduction

Include:

- Problem background
- Motivation
- Stakeholders/users
- Why robotics is appropriate
- Brief description of solution

End with:

“This paper proposes….”

Problem Definition & Requirements

A. List measurable goals - Metric & target

B. Constraints & Assumptions

Discuss:

- Cost
- Environment
- Safety
- Energy
- Size/weight
- Regulations

C. Engineering Requirements

D. Convert goals into technical specs.

E. System Architecture & Methodology (1.5–2 pages)

F. Overall Architecture

Insert system block diagram

(Must show sensors → controller → actuators → power → communication)

Put in steps

3. Subsystem Breakdown –

Explain each subsystem and its function

4. Design Methodology

Explain engineering process:

- Concept selection
- Trade-offs
- Decision logic

5. Robotics Design & Technical Implementation (≈ 2–2.5 pages)

A. Sensors

Explain: Type, Specification, Justification

B. Actuators/Mechanics

Explain: Mobility, manipulation, speed, load reasoning

C. Control Strategy

Explain: Control logic, algorithms, Flowcharts or equations

D. Power & Energy System

Include: Battery choice, Energy estimate, Runtime calculations, Power supply with reason

E. Safety & Reliability

Discuss: Hazards, Protections, Fail-safes

6. Modelling, Analysis & Validation (≈ 1.5–2 pages)

Include quantitative work:

Examples:

- Coverage/time calculations
- Energy consumption estimates
- Speed/path planning
- Accuracy or detection analysis
- Simulations

A. Calculations - Show equations/estimates.

B. Simulation/Testing - Include graphs/tables.

C. Results Interpretation - Explain what results mean.

7. Integration & Feasibility (0.5–1 page)

Explain:

- How subsystems connect
- Implementation plan
- Deployment scenario
- Risks & limitations
- Mitigation strategies

8. Sustainability & SDG Impact (1 page)

A. SDG Alignment

Explain - Which SDG target, How solution directly addresses it

B. Environmental Impact

Discuss - Energy efficiency, Resource usage, Waste reduction

C. Social & Ethical Considerations - Include quantification if possible.

Discuss - Accessibility, Cost, Safety, Community benefit

9. Conclusion (0.5 page)

Summarize - Key design contributions, Expected impact, Feasibility, Future improvements

References (IEEE format)

[1] Author, “Title,” Journal/Website, Year.
[2]Use authoritative sources only.

Appendices (Optional – NOT included in page count)

Include:

- Extra calculations
- Large diagrams
- Code snippets
- Additional simulations